

Three-dimensional saturated-unsaturated flow with axial symmetry to a partially penetrating well in a compressible unconfined aquifer

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[1] A new analytical solution is presented for the delayed response process characterizing flow to a partially penetrating well in an unconfined aquifer. The new solution generalizes that of Neuman (1972, 1974) by accounting for unsaturated flow above the water table. Three-dimensional, axially symmetric flow in the unsaturated zone is described by a linearized version of Richards' equation in which hydraulic conductivity and water content vary exponentially with incremental capillary pressure head relative to its air entry value (defining the interface between the saturated and unsaturated zones). Unsaturated soil properties are characterized by an exponent κ having the dimension of inverse length or, equivalently, a dimensionless exponent $\kappa_D = \kappa b$, where b is initial saturated thickness. Our treatment of the unsaturated zone is similar to that of Kroszynski and Dagan (1975), who, however, have ignored internal (artesian) aquifer storage. According to Kroszynski and Dagan, aquifers that are not excessively shallow have values of κ_D (their parameter a) much greater than 10. We find that in such typical cases, unsaturated flow has little impact on early and late dimensionless time drawdown a short distance below the water table. Unsaturated flow causes drawdown to increase slightly at intermediate dimensionless time values that represent transition from an early artesian-dominated to a late water-table-dominated flow regime. Delayed drainage from the unsaturated zone becomes less and less important as κ_D increases; as $\kappa_D \rightarrow \infty$, this effect dies out, and drawdown is controlled entirely by delayed decline in the water table as in the model of Neuman. The unsaturated zone has a major impact on drawdown at intermediate time and a significant impact at early and late times, in the atypical case of $\kappa_D \leq 1$, becoming the dominant factor as κ_D approaches zero (the soil water retention capacity becomes very large and/or saturated thickness becomes insignificant). Our new solution was used to analyze field data from a pumping test conducted by Moench et al. (2001) in a glacial outwash deposit at Cape Cod, Massachusetts. The solution was fitted individually and simultaneously to time-drawdown data from 20 piezometers and observation wells and simultaneously to data from three piezometers in each of two clusters at various depths and distances from the pumping well, with very good results. Our parameter estimates of hydraulic conductivities from the simultaneous fit are similar to those obtained previously by Moench (2004), but our estimates of specific yield and storage are smaller and larger, respectively, while our estimate of κ is not comparable with his estimates of three empirical parameters.

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1. Introduction

[2] In analyzing constant rate pumping tests in unconfined aquifers one often finds that drawdown fails to follow the classical solutions of *Theis* [1935] for fully penetrating wells or of *Hantush* [1964] for partially penetrating wells. This is not surprising considering that both solutions were

intended for confined aquifers. A plot of log drawdown versus log time in an observation well or a piezometer completed in an unconfined aquifer often delineates an inflected S-shaped curve consisting of a steep early segment, a flat segment at intermediate times, and a somewhat steeper segment at later times. The early segment implies that some water is released from aquifer storage instantaneously in response to head drops occurring shortly after pumping starts. The intermediate segment suggests the presence of an additional source of water released from storage following some time delay, a phenomenon known as delayed yield or delayed response. The late segment represents the combined effect of both storage mechanisms.

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[3] A mathematical model of delayed yield was originally proposed by *Boulton* [1955, 1963] for fully penetrating wells. In his model, Boulton assumed that flow takes place horizontally and the amount of water released from storage per unit horizontal area of an aquifer, due to a unit drop in head occurring at time τ , is the sum of two components: S , a volume of water released instantaneously at time τ from artesian (compressible) storage in the aquifer interior and S_y , a volume the release of which is delayed in time, decaying exponentially as $\alpha S_y \exp[-\alpha(t - \tau)]$ where t is time ($t \geq \tau$) and α is a characteristic constant of the aquifer. Two closely related models were reported later by *Boulton* [1970] and *Boulton and Pontin* [1971]. The original theory was used by *Boulton* [1963] and *Prickett* [1965] to propose a graphical procedure for evaluating the transmissivity T , storativity S , delayed storage parameter S_y (commonly taken to represent specific yield or drainable porosity), and delay index $1/\alpha$ of an unconfined aquifer by matching time-drawdown data to theoretical type curves.

[4] Though his exponential release model was based on a conjecture (being thus semiempirical), Boulton postulated three possible mechanisms for such delayed release: leakage from either a bed of fine-grained material overlying the aquifer and containing the water table or from fine-grained compressible layers interbedded with the aquifer material [*Boulton*, 1955], or delayed release from storage above the water table [*Boulton*, 1963]. Boulton's idea of delayed yield from fine-grained interbeds is conceptually and mathematically equivalent to that of a dual porosity model for flow through fractured rocks [*Barenblatt et al.*, 1960; *Warren and Root*, 1963], the latter having a well-defined mechanistic basis in which α is explicitly related to the permeability and physical dimensions of a typical rock matrix block [e.g., *Gringarten*, 1982, 1984]. *Bower and Rice* [1978] conjectured that delayed yield could be related to delayed air entry into the unsaturated zone immediately above the water table.

[5] A mechanistic model reproducing the intermediate and late time drawdown behavior of unconfined aquifers was proposed for fully penetrating wells by *Boulton* [1954] and for partially penetrating wells by *Dagan* [1967]. This model treats the water table as a moving material boundary of a rigid saturated flow domain in which flow can take place both horizontally and vertically under the combined actions of differential pressure and gravity. Because of the assumption of incompressibility, flow is governed by Laplace's equation, which causes drawdowns to exhibit a step increase (jump) when pumping starts. This is in contrast to the steep but gradual increase observed in nature and predicted by *Boulton's* [1955, 1963] delayed yield model, owing to its consideration of artesian storage in the unconfined aquifer interior through inclusion of the compressive storativity S .

[6] A mechanistic model accounting for artesian storage, which treats the water table as a moving boundary of a compressible saturated domain in which flow takes place both horizontally and vertically, was proposed for fully penetrating wells by *Neuman* [1972, 1973] and for partially penetrating wells by *Neuman* [1974]. In *Neuman's* model flow is governed by a diffusion equation, which allows reproducing all three segments of a typical S-shaped time-drawdown curve; instead of referring to delayed yield above

the water table as does *Boulton* [1963], *Neuman* [1972, 1974] refers to delayed water table decline in response to vertical hydraulic gradients that evolve gradually (rather than exhibiting a jump) in the underlying saturated zone. In the case of fully penetrating wells, *Neuman's* [1975] model gives rise to type curves resembling closely those produced by *Prickett* [1965] on the basis of *Boulton's* [1963] delayed yield model. In fact, the two sets of type curves virtually coincide if one redefines *Boulton's* α as [*Neuman*, 1975]

$$\alpha = \frac{K_z}{S_y b} [3.06 - 0.57 \log(K_D r_D^2)] \quad (1)$$

where S_y is a constant specific yield (drainable porosity); $K_D = K_z/K_r$ is the saturated hydraulic anisotropy ratio, K_z and K_r being the vertical and horizontal saturated hydraulic conductivities, respectively; b is the initial saturated thickness; and $r_D = r/b$ is normalized radial distance r from the pumping well. According to (1) α does not reflect unsaturated flow, suggesting that the same may be true for *Boulton's* [1963] delayed yield model. Our opinion that this latter model may be viewed as representative of saturated flow at and below the water table, not of unsaturated flow above the water table as intimated by Boulton (and held by many), is supported further by an integrodifferential representation [*Herrera et al.*, 1978] of the same flow mechanism as that considered by *Neuman* [1972, 1975], which yields a series expansion for α with a leading term $3K_z/S_y b$ and a remainder that varies with r_D , as does the logarithmic term in (1). Our opinion is further reinforced by the mechanistic model of *Streletsova* [1972a, 1972b, 1973], which is similar in principle to that of *Neuman* [1972] except for her reliance on a simplifying assumption that drawdown s_{wt} at the water table increases at a rate $(3K_z/S_y b)(\bar{s} - s_{wt})$ where $\bar{s}$ is vertically averaged drawdown. *Streletsova's* model yields a mathematical expression for $\bar{s}$ that coincides with that of *Boulton* [1963] provided only that one sets $\alpha = 3K_z/S_y b$, the leading term in the corresponding expressions of *Neuman* [1975] (our equation (1)) and *Herrera et al.* [1978].

[7] *Kroszynski and Dagan* [1975] were the first to account analytically for unsaturated flow above the water table on drawdown at and below the water table when a well is pumped at a constant rate. Their solution, which considers partially penetrating wells, is an extension of *Dagan's* [1967] in that it treats the saturated zone as being rigid and consequently predicts a step increase in drawdown below the water table when pumping starts. *Moench* [1995, 1997] overcame this last difficulty by adopting *Neuman's* [1972, 1973, 1974] saturated flow model with a modified boundary condition at the water table, accounting additionally for borehole storage. In both models the water table remains, to a leading order of approximation, at its original position. Whereas in *Neuman's* model the amount of water released from storage per unit horizontal area of an aquifer, due to a unit drop in head at the water table occurring at time τ , is equal to the specific yield S_y , in *Moench's* [1995] model this amount is released from storage at an exponentially decaying rate $\alpha S_y \exp[-\alpha(t - \tau)]$, $t \geq \tau$, where in his view α is a characteristic constant of the aquifer, representing vertical flow through an unsaturated column above the water table. *Moench* considered the

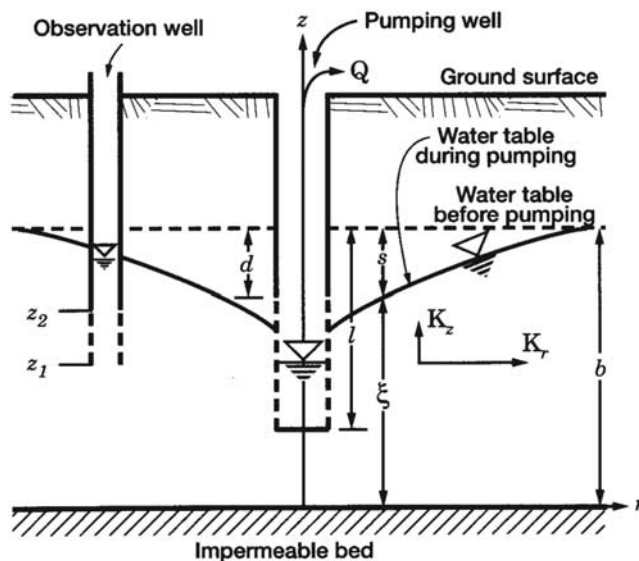


Figure 1. Schematic representation of system geometry.

“theoretical basis” for his model to be “weak” (an observation confirmed by *Narasimhan and Zhu* [1993]), its underlying exponential decay rate being “nothing more than... empirical.” This notwithstanding, *Moench* [1994] was able to fit drawdowns measured during three pumping tests in mildly heterogeneous aquifers (at Saint Pardon de Conques, France [*Neuman*, 1975]; Borden, Ontario, Canada [*Nwankwor et al.*, 1984]; and Cape Cod, Massachusetts [*Moench et al.*, 2001]) more closely to his than to *Neuman’s* [1974] theoretical solution by analyzing data from several wells jointly. Significantly, his solution [*Moench*, 1995] yielded a more realistic estimate of specific yield than did that of *Neuman* [1974]. Further improvement in the fit of Cape Cod drawdown data to theoretical type curves was obtained by *Moench et al.* [2001] and *Moench* [2004] upon considering a sum of exponential decay terms with three fitting parameters $\alpha_1, \alpha_2, \alpha_3$ instead of only one.

[8] In this paper we present a new analytical solution generalizing that of *Neuman* [1974] by accounting for unsaturated flow above the water table. Whereas our treatment of the saturated zone is identical to that of *Neuman* (who took into account aquifer compressibility but not unsaturated flow), our handling of the unsaturated zone is similar to that of *Kroszynski and Dagan* [1975] (who did the opposite). The new solution describes fully coupled three-dimensional flow with axial symmetry through both the saturated and the unsaturated zones. Defining the water table as an isobar at air entry pressure introduces a saturated capillary fringe at nonpositive pressures between this and an underlying zero-pressure isobar, the latter forming a traditional water table below which pressures are positive. Flow in the saturated zone (including the capillary fringe) is governed by a linear diffusion equation and that above the capillary fringe by a linearized version of Richards' equation. Such linearization is enabled by expressing hydraulic conductivity and water content as exponential functions of incremental capillary pressure head relative to its air entry value. This allows characterizing the unsaturated zone by an exponent κ having the dimension of inverse length and a dimensionless exponent $\kappa_D = \kappa b$ where b is initial saturated

thickness. We explore the effect of κ_D on aquifer behavior and use our new solution to reanalyze the Cape Cod pumping test data of *Moench et al.* [2001].

[9] Concurrently with our work, *Mathias and Butler* [2006] proposed a related solution which however is limited to one-dimensional vertical flow through the unsaturated zone. Whereas we employ a single κ to characterize both hydraulic conductivity and water content as functions of pressure head, and consider the unsaturated zone to have infinite thickness, they allow characterizing each of these two functions by means of a different exponent and treat the unsaturated zone as having finite thickness. Whereas *Mathias and Butler* consider a pumping well of zero radius that fully penetrates the aquifer, they state that their solution is easily extended to the case of a partially penetrating well with storage. Our solution can likewise be extended to include two separate exponents, finite unsaturated zone thickness and borehole storage. We however demonstrate on the basis of our existing solution that (1) flow in the unsaturated zone is far from one-dimensional around the pumping well and (2) a single value of κ with infinite unsaturated zone thickness and a pumping well of zero radius can be used parsimoniously and effectively to interpret, individually as well as simultaneously, time-drawdown data from all 20 piezometers and observational wells during the Cape Cod pumping test of *Moench et al.* [2001]. As *Mathias and Butler* [2006] have interpreted only one time-drawdown record from a single observation well at Cape Cod while ignoring partial penetration, which dominates drawdowns at this site, the validity and generality of their interpretation remains in question.

2. Theory

2.1. Problem Definition

[10] We consider a compressible unconfined aquifer of infinite lateral extent resting on an impermeable boundary (Figure 1). The aquifer is spatially uniform and anisotropic with a fixed ratio $K_D = K_z/K_r$ between the vertical and horizontal saturated hydraulic conductivities, K_z and K_r , respectively. The aquifer is saturated beneath an initially horizontal water table at elevation $z = b$ above the bottom, defined as a $\psi = \psi_a$ isobar where ψ is pressure head and $\psi_a \leq 0$ is the pressure head required for air to enter a saturated medium. A saturated capillary fringe at nonpositive pressure $\psi_a \leq \psi \leq 0$ extends from the water table down to the $\psi = 0$ isobar (traditional water table) at elevation $b + \psi_a$ (note that the capillary fringe disappears if one sets $\psi_a = 0$). Prior to the onset of pumping the saturated and overlying unsaturated zones are at uniform initial hydraulic head $h_0 = b + \psi_a$. Starting at time $t = 0$, water is withdrawn at a constant volumetric rate Q from a well of zero radius that penetrates the saturated zone between depths l and d below the initial water table (at air entry pressure). Under these conditions the drawdown $s(r, z, t) = h_0 - h(r, z, t)$ in the saturated zone is governed by the diffusion equation

$$K_r \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial s}{\partial r} \right) + K_z \frac{\partial^2 s}{\partial z^2} = S_s \frac{\partial s}{\partial t} \quad 0 \leq z < \xi \quad (2)$$

subject to the initial condition

$$\xi(r, 0) = b \quad \text{or} \quad s(r, z, 0) = s_0 = 0 \quad (3)$$

and boundary conditions

$$s(\infty, z, t) = 0 \quad (4)$$

$$\frac{\partial s}{\partial z} = 0 \quad z = 0 \quad (5)$$

$$\lim_{r \rightarrow 0} r \frac{\partial s}{\partial r} = 0 \quad 0 \leq z \leq b-l \quad b-d \leq z \leq \xi \quad (6)$$

$$\lim_{r \rightarrow 0} \int_{b-l}^{\min(b-d, \xi)} r \frac{\partial s}{\partial r} dz = -\frac{Q}{2\pi K_r} \quad b-l \leq z \leq b-d \quad (7)$$

where S_s is specific storage and $\xi = b - s(r, \xi, t)$ is head at the water table. If the top of the unsaturated zone is at elevation $z = b + L$ (L being the thickness of the unsaturated zone) then the drawdown $\sigma(r, z, t) = h_0 - h(r, z, t) = b + \psi_a - h(r, z, t)$ in this zone is controlled by Richards' equation

$$K_r \frac{1}{r} \frac{\partial}{\partial r} (k(\psi) r \frac{\partial \sigma}{\partial r}) + K_z \frac{\partial}{\partial z} (k(\psi) \frac{\partial \sigma}{\partial z}) = C(\psi) \frac{\partial \sigma}{\partial t} \quad \xi < z < b+L \quad (8)$$

subject to the initial condition

$$\sigma_0 = \sigma(r, z, 0) = 0 \quad (9)$$

and boundary conditions

$$\sigma(\infty, z, t) = 0 \quad (10)$$

$$\frac{\partial \sigma}{\partial z} = 0 \quad z = b+L \quad (11)$$

$$\lim_{r \rightarrow 0} \frac{\partial \sigma}{\partial r} = 0 \quad \xi < z < b+L \quad (12)$$

where $0 \leq k(\psi) \leq 1$ is relative (ratio of actual to saturated) hydraulic conductivity and $C(\psi) \geq 0$ is specific moisture capacity

$$C(\psi) = d\theta/d\psi, \quad (13)$$

θ being volumetric water content. The unsaturated and saturated flow regimes are coupled by interface conditions representing continuity of pressure and normal flux across the water table,

$$s = \sigma \quad z = \xi \quad (14)$$

$$\nabla s \cdot \mathbf{n} = \nabla \sigma \cdot \mathbf{n}, \quad z = \xi \quad (15)$$

where $\nabla = (\partial/\partial r, \partial/\partial z)^T$ is the three-dimensional axially symmetric gradient operator, the superscript T denoting transpose, and $\mathbf{n}$ is a unit vector normal to the water table.

2.2. Linearization

[11] The above system of equations is highly nonlinear because of (1) the nonlinear nature of Richards' equation and (2) the presence of a moving interface (water table) between two different (saturated and unsaturated) flow regimes. To solve it we restrict ourselves to a pumping rate

Q that is small compared to $K_r b^2$, expand the dependent variable in power series

$$s = s^{(0)} + Qs^{(1)} + Q^2s^{(2)} + \dots \quad s^{(0)} = 0$$

$$\sigma = \sigma^{(0)} + Q\sigma^{(1)} + Q^2\sigma^{(2)} + \dots \quad \sigma^{(0)} = 0$$

$$\psi = \psi^{(0)} + Q\psi^{(1)} + Q^2\psi^{(2)} + \dots \quad \psi^{(0)} = b + \psi_a - z$$

$$\theta = \theta^{(0)} + Q\theta^{(1)} + Q^2\theta^{(2)} + \dots \quad \theta^{(0)} = \theta(z, 0) \text{ corresponding to } \psi^{(0)}$$

$$\xi = \xi^{(0)} + Q\xi^{(1)} + Q^2\xi^{(2)} + \dots \quad \xi^{(0)} = b$$

(16)

and disregard terms of order higher than first in Q , as did *Kroszynski and Dagan* [1975]. Substituting the remaining linear terms (in Q) into (2)–(15) and treating horizontal flux into the pumping well as if it was vertically uniform (it generally is not, but was assumed to be such by *Hantush* [1964], *Dagan* [1967], *Neuman* [1974] and *Moench* [1995, 1997], introducing some errors near the edges of the pumping well) leads to the following first-order representation of these equations for the saturated zone,

$$K_r \frac{1}{r} \frac{\partial}{\partial r} (r \frac{\partial s}{\partial r}) + K_z \frac{\partial^2 s}{\partial z^2} = S_s \frac{\partial s}{\partial t} \quad 0 \leq z < b \quad (17)$$

$$\xi(r, 0) = b \quad \text{or} \quad s_0 = 0 \quad (18)$$

$$s(\infty, z, t) = 0 \quad (19)$$

$$\frac{\partial s}{\partial z} = 0 \quad z = 0 \quad (20)$$

$$\lim_{r \rightarrow 0} r \frac{\partial s}{\partial r} = 0 \quad 0 \leq z \leq b-l \quad b-d \leq z \leq b \quad (21)$$

$$\lim_{r \rightarrow 0} r \frac{\partial s}{\partial r} = -\frac{Q}{2\pi K_r(l-d)} \quad b-l \leq z \leq b-d. \quad (22)$$

The corresponding first-order (linearized) unsaturated flow equations are

$$K_r k_0(z) \frac{1}{r} \frac{\partial}{\partial r} (r \frac{\partial \sigma}{\partial r}) + K_z \frac{\partial}{\partial z} (k_0(z) \frac{\partial \sigma}{\partial z}) = C_0(z) \frac{\partial \sigma}{\partial t} \quad b < z < b+L \quad (23)$$

$$k_0(z) = k(\theta_0) \quad C_0(z) = C(\theta_0) \quad (24)$$

$$\sigma_0 = 0 \quad (25)$$

$$\sigma(\infty, z, t) = 0 \quad (26)$$

$$\frac{\partial \sigma}{\partial z} = 0 \quad z = b+L \quad (27)$$

$$\lim_{r \rightarrow 0} r \frac{\partial \sigma}{\partial r} = 0 \quad b < z < b+L. \quad (28)$$

The linearized interface conditions at the water table are

$$s - \sigma = 0 \quad z = b \quad (29)$$

$$\frac{\partial s}{\partial z} - \frac{\partial \sigma}{\partial z} = 0 \quad z = b. \quad (30)$$

[12] Like *Kroszynski and Dagan* [1975], we represent unsaturated medium properties by the exponential constitutive relationships $k(\psi) = e^{\kappa(\psi - \psi_a)}$, $\kappa > 0$ [Gardner, 1958] and $\theta(\psi) - \theta_r = S_y e^{\kappa(\psi - \psi_a)}$ where θ_r is residual water content and $S_y = \theta_{sat} - \theta_r$ is drainable porosity or specific yield. It follows that $C(\psi) = S_y d e^{\kappa(\psi - \psi_a)} / d\psi = S_y \kappa e^{\kappa(\psi - \psi_a)}$, $k_0(z) = e^{\kappa(b-z)}$ and $C_0(z) = S_y \kappa e^{\kappa(b-z)}$.

2.3. Solution in Time and Laplace-Transformed Domains

[13] In a manner analogous to *Neuman* [1974] we decompose s into two parts (Appendix A),

$$s = s_H + s_U \quad (31)$$

where s_H is *Hantush's* [1964, pp. 347–350] solution for a partially penetrating well in a confined aquifer and s_U is a correction due to saturated-unsaturated unconfined conditions. *Hantush's* solution further decomposes into

$$s_H = s_T + \Delta s, \quad (32)$$

where s_T is the *Theis* [1935] solution

$$s_T = \frac{Q}{4\pi T} \int_{1/4t_s}^{\infty} \frac{e^{-y}}{y} dy \quad (33)$$

and Δs is a correction due to partial penetration given by

$$\Delta s = \frac{Q}{4\pi T} \frac{2}{\pi(l_D - d_D)} \sum_{n=1}^{\infty} \frac{1}{n} \cos[n\pi(1 - z_D)] \cdot [\sin(n\pi l_D) - \sin(n\pi d_D)] \int_{1/4t_s}^{\infty} \exp\left[-y - \frac{\beta(n\pi)^2}{4y}\right] \frac{dy}{y} \quad (34)$$

in which $T = K_r b$ is transmissivity, $t_s = \alpha_s t / r^2$ is dimensionless time with respect to the compressive hydraulic diffusivity $\alpha_s = K_r / S_s$, $l_D = l/b$, $d_D = d/b$, $z_D = z/b$ and $\beta = K_D r_D^2$, $K_D = K_z / K_r$ being saturated hydraulic anisotropy ratio and $r_D = r/b$.

[14] We show in Appendix B that for an unsaturated zone of infinite thickness ($L \rightarrow \infty$), the Laplace transforms of s_U and σ are given by

$$\bar{s}_U(r_D, z_D, p_D) = \frac{Qt}{2\pi T p_D} \int_0^{\infty} \frac{\cosh(\mu z_D) \left(\frac{\kappa_D}{2} - \sqrt{\frac{\kappa_D^2}{4} + \nu^2} \right)}{\cosh(\mu) \left(\frac{\kappa_D}{2} - \sqrt{\frac{\kappa_D^2}{4} + \nu^2} \right) - \mu \sinh(\mu)} \cdot \frac{\sinh[\mu(1 - l_D)] - \sinh[\mu(1 - d_D)]}{\mu^2(l_D - d_D) \sinh(\mu)} y J_0(y \beta^{1/2}) dy \quad (35)$$

$$\bar{\sigma}(r_D, z_D, p_D) = \frac{Qt}{2\pi T p_D} \int_0^{\infty} \frac{\eta \sinh(\eta)}{\cosh(\eta) \left(\frac{\kappa_D}{2} - \sqrt{\frac{\kappa_D^2}{4} + \nu^2} \right) - \eta \sinh(\eta)} \cdot \exp\left[\left(\frac{\kappa_D}{2} - \sqrt{\frac{\kappa_D^2}{4} + \nu^2}\right)(z_D - 1)\right] \cdot \frac{\sinh[\eta(1 - l_D)] - \sinh[\eta(1 - d_D)]}{\eta^2(l_D - d_D) \sinh(\eta)} y J_0(y \beta^{1/2}) dy \quad (36)$$

respectively where $\mu^2 = y^2 + \frac{p_D}{\kappa_s \beta}$, $\nu^2 = y^2 + \frac{p_D}{\kappa_s \phi \beta}$, $\phi = \frac{S}{\kappa_D S_y}$ and $p_D = tp$, p being the parameter of the Laplace transform defined in (B2).

2.4. Numerical Inversion of Laplace Transform

[15] The inverse Laplace transform of (35) is given formally by Mellin's inversion formula [e.g., *Carlslaw and Jaeger*, 1963] as

$$s_U(r_D, z_D, t_s) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} (e^{p_D} - 1) \frac{\bar{s}_U(r_D, z_D, p_D)}{t} dp_D \quad (37)$$

where $\gamma > 0$ is a real constant and p_D is now a complex variable. Assuming that the order of integration with respect to p_D and y is interchangeable, (37) can be rewritten as

$$s_U(r_D, z_D, t_s) = \frac{Q}{2\pi T} \int_0^{\infty} f(r_D, z_D, t_s; y) y J_0(y \beta^{1/2}) dy \quad (38)$$

where $f(r_D, z_D, t_s; y)$ is the inverse Laplace transform of

$$\bar{f}(r_D, z_D, p_D; y) = \frac{e^{p_D} - 1}{p_D} \frac{\left(\frac{\kappa_D}{2} - \sqrt{\frac{\kappa_D^2}{4} + \nu^2} \right) \cosh(\mu z_D)}{\left[\cosh(\mu) \left(\frac{\kappa_D}{2} - \sqrt{\frac{\kappa_D^2}{4} + \nu^2} \right) - \mu \sinh(\mu) \right]} \cdot \frac{\sinh[\mu(1 - l_D)] - \sinh[\mu(1 - d_D)]}{\mu^2(l_D - d_D) \sinh(\mu)} \quad (39)$$

Similarly, the inverse Laplace transform of (36) is given by

$$\sigma(r_D, z_D, t_s) = \frac{Q}{2\pi T} \int_0^{\infty} g(r_D, z_D, t_s; y) y J_0(y \beta^{1/2}) dy \quad (40)$$

where $g(r_D, z_D, t_s; y)$ is the inverse Laplace transform of

$$\bar{g}(r_D, z_D, p_D; y) = \frac{e^{p_D} - 1}{p_D} \frac{\eta \sinh(\eta)}{\cosh(\eta) \left(\frac{\kappa_D}{2} - \sqrt{\frac{\kappa_D^2}{4} + \nu^2} \right) - \eta \sinh(\eta)} \cdot \frac{\sinh[\eta(1 - l_D)] - \sinh[\eta(1 - d_D)]}{\eta^2(l_D - d_D) \sinh(\eta)} \cdot \exp\left[\left(\frac{\kappa_D}{2} - \sqrt{\frac{\kappa_D^2}{4} + \nu^2}\right)(z_D - 1)\right] \quad (41)$$

[16] We evaluate $f(r_D, z_D, t_s; y)$ and $g(r_D, z_D, t_s; y)$ by numerical inversion of $\bar{f}(r_D, z_D, p_D; y)$ and $\bar{g}(r_D, z_D, p_D; y)$, respectively, using an algorithm by *Crump* [1976] as modified by *de Hoog et al.* [1982]. Whereas standard inversion with respect to p is done over a time interval $[0, t]$, we do the inversion with respect to p_d over a unit dimensionless time (defined as p_D^{-1}) interval $[0, 1]$, regardless of what t_s is.

2.5. Vertically Averaged Drawdown in an Observation Well

[17] Drawdown in a piezometer or an observation well that penetrates the saturated zone between elevations z_1/b

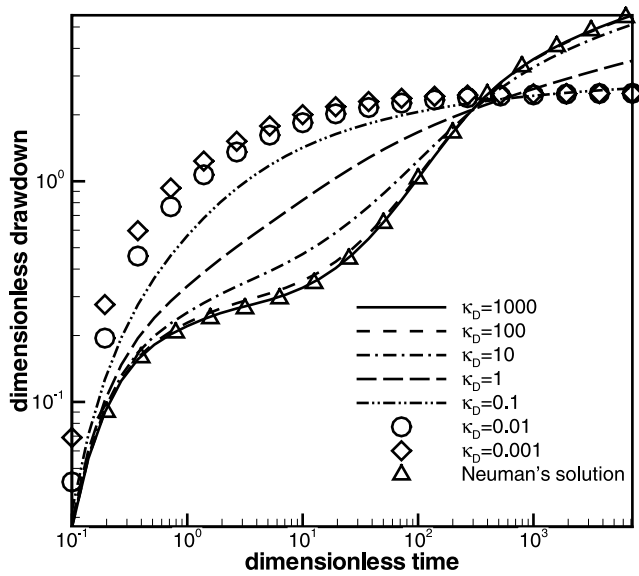


Figure 2. Dimensionless drawdown, $s_D = 4\pi Ts/Q$, versus dimensionless time, $t_s = Tt/Sr^2$, at $r/b = 0.5$, $z/b = 0.8$ due to pumping from a well penetrating the upper 20% of the saturated zone of an aquifer characterized by $K_z/K_r = 1$, $S/S_y = 0.01$ and various values of $\kappa_D = \kappa b$. *Neuman's* [1974] solution is included for reference.

and z_2/b (Figure 1) is obtained by averaging the above point drawdown over this interval,

$$s_{z_2-z_1}(r_D, t_s) = \frac{1}{z_2/b - z_1/b} \int_{z_1/b}^{z_2/b} s(r_D, z_D, t_s) dz_D. \quad (42)$$

3. Effect of Unsaturated Flow on Drawdown

[18] According to our solution the effect of unsaturated flow on drawdown is captured fully by the dimensionless parameter $\kappa_D = \kappa b$ in which κ is a constitutive exponent having the dimension of inverse length and b is the initial saturated thickness. To investigate the manner in which κ_D impacts drawdown in the saturated zone, we start by considering an isotropic aquifer ($K_D = 1$) with a ratio $S/S_y = 1/100$ between artesian storativity and specific yield. Figure 2 shows how log dimensionless drawdown $s_D = 4\pi K_r b s/Q$ varies with log dimensionless time $t_s = K_r b t/(Sr^2)$ at dimensionless elevation $z/b = 0.8$ and dimensionless radial distance $r/b = 0.5$ from a pumping well penetrating the upper 20% of the (initial) saturated thickness. Unsaturated flow is seen to have significant impact on dimensionless log-log time-drawdown behavior when κ_D is less than 1 (the aquifer has large retention capacity and/or small initial saturated thickness). The impact of unsaturated flow diminishes as κ_D increases, becoming small or insignificant at κ_D greater than 10 (small retention capacity and/or large saturated thickness). At these large κ_D values, our solution practically coincides with that of *Neuman* [1974].

[19] When κ_D is small, unsaturated flow causes dimensionless drawdown to increase (relative to that predicted by *Neuman* [1974]) at early and intermediate dimensionless times and to decrease at late time, the impact being least pronounced at early time and most pronounced at interme-

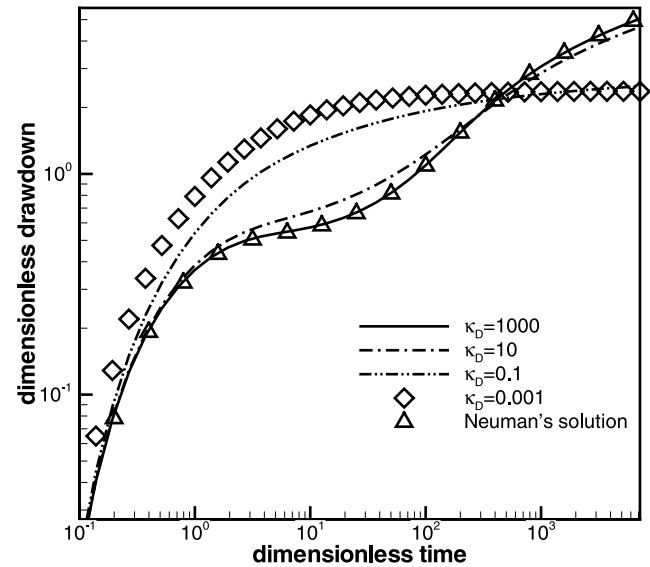


Figure 3. Dimensionless drawdown, $s_D = 4\pi Ts/Q$, versus dimensionless time, $t_s = Tt/Sr^2$, at $r/b = 0.5$, $z/b = 0.8$ due to pumping from a well fully penetrating the saturated zone of an aquifer characterized by $K_z/K_r = 1$, $S/S_y = 0.01$, and various values of $\kappa_D = \kappa b$. *Neuman's* [1972, 1973] solution is included for reference.

diate time (Figure 2). The overall effect is to eliminate S-shaped inflections, rendering the dimensionless log-log time-drawdown curve more monotonic. Whereas S-shaped behavior at large κ_D reflects the dominance of artesian storage during early time and of near instantaneous drainage at the water table during late time, monotonic behavior at

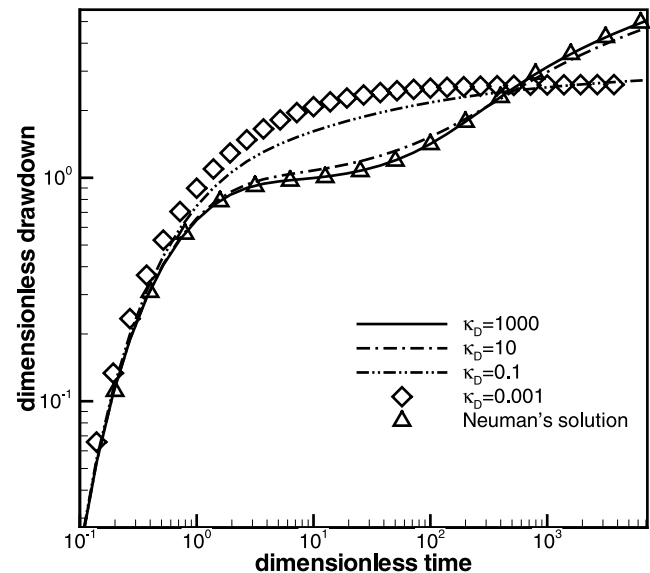


Figure 4. Dimensionless drawdown, $s_D = 4\pi Ts/Q$, versus dimensionless time, $t_s = Tt/Sr^2$, in a fully penetrating observation well at $r/b = 0.5$ due to pumping from a well fully penetrating the saturated zone of an aquifer characterized by $K_z/K_r = 1$, $S/S_y = 0.01$, and various values of $\kappa_D = \kappa b$. *Neuman's* [1972, 1973] solution is included for reference.

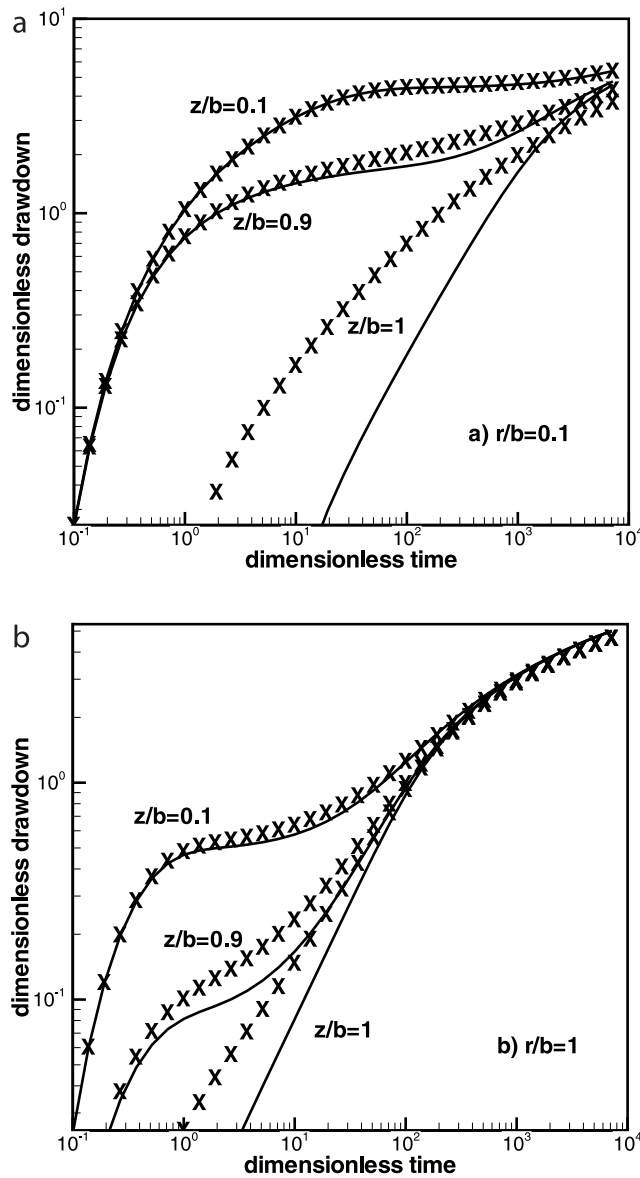


Figure 5. Dimensionless drawdown, $s_D = 4\pi Ts/Q$, versus dimensionless time, $t_s = Tt/Sr^2$, at (a) $r/b = 0.1$ and (b) $r/b = 1.0$, $z/b = 0.1, 0.9, 1.0$ due to pumping from a well fully penetrating the saturated zone of an aquifer characterized by $K_z/K_r = 1$, $S/S_y = 0.01$, and $\kappa_D = 10$ (crosses), $\kappa_D = 1000$ (solid curves).

small κ_D reflects the dominance of slow drainage from the unsaturated zone at all times.

[20] Figure 3 depicts dimensionless log-log time-drawdown behavior under conditions similar to those in Figure 2 but for a fully penetrating pumping well. Comparing Figures 2 and 3 reveals that the effect of unsaturated flow on drawdown in the saturated zone diminishes as the degree of penetration increases. The reason is that as the pumping well penetrates the saturated zone more fully vertical flux components, which dominate drainage from the unsaturated zone, diminish. Comparing Figure 3 with Figure 4, which represents vertically averaged drawdown in a fully penetrating observation well due to pumping from a fully penetrating well, indicates that such averaging like-

wise causes the impact of unsaturated flow on dimensionless drawdown in the saturated zone to decrease. This is so because the average is less sensitive than are point drawdowns to vertical fluxes and hence to drainage from above the water table.

[21] Figure 5 compares time-drawdown behaviors at dimensionless elevations $z/b = 0.1, 0.9$ and 1.0 at dimensionless distances $r/b = 0.1$ and 1 from a fully penetrating pumping well corresponding to $\kappa_D = 10$ and 1000 . The effect of unsaturated flow is most pronounced at the water table, more so closer to the pumping well than farther from it, and least near the aquifer bottom. A decrease in κ_D leads to an increase in point drawdown. Rendering the aquifer anisotropic with $K_z = 0.1 K_r$ (Figure 6) reduces the unsat-

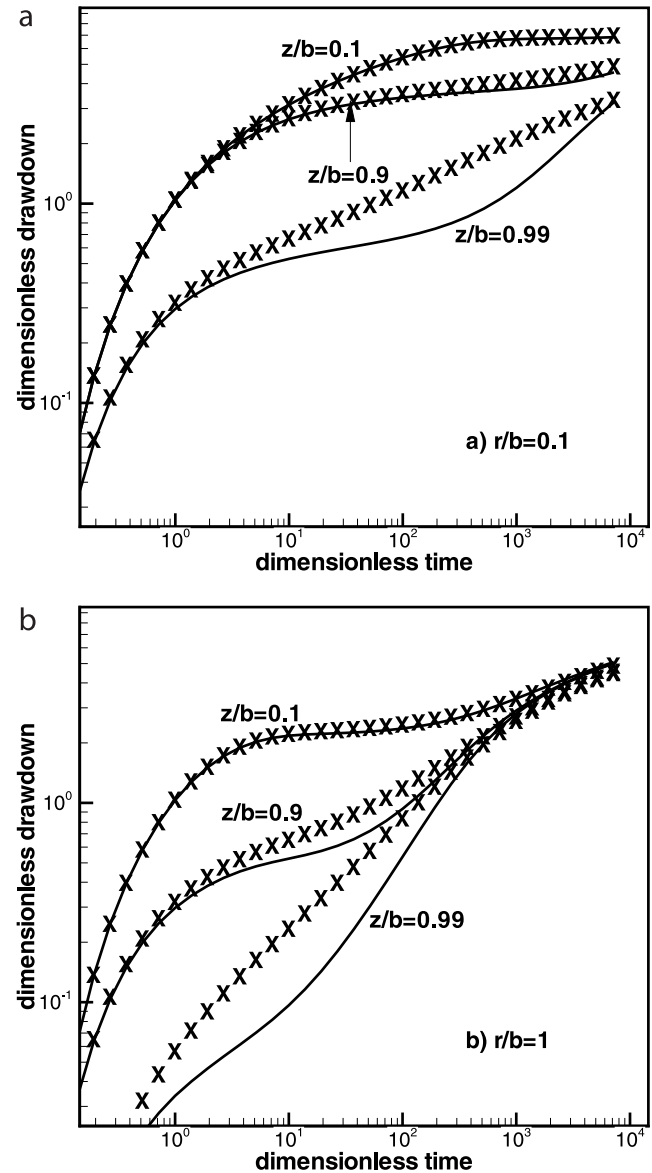


Figure 6. Dimensionless drawdown, $s_D = 4\pi Ts/Q$, versus dimensionless time, $t_s = Tt/Sr^2$, at (a) $r/b = 0.1$ and (b) $r/b = 1.0$, $z/b = 0.1, 0.9, 0.99$ due to pumping from a well fully penetrating the saturated zone of an aquifer characterized by $K_z/K_r = 0.1$, $S/S_y = 0.01$, and $\kappa_D = 10$ (crosses), $\kappa_D = 1000$ (solid curves).

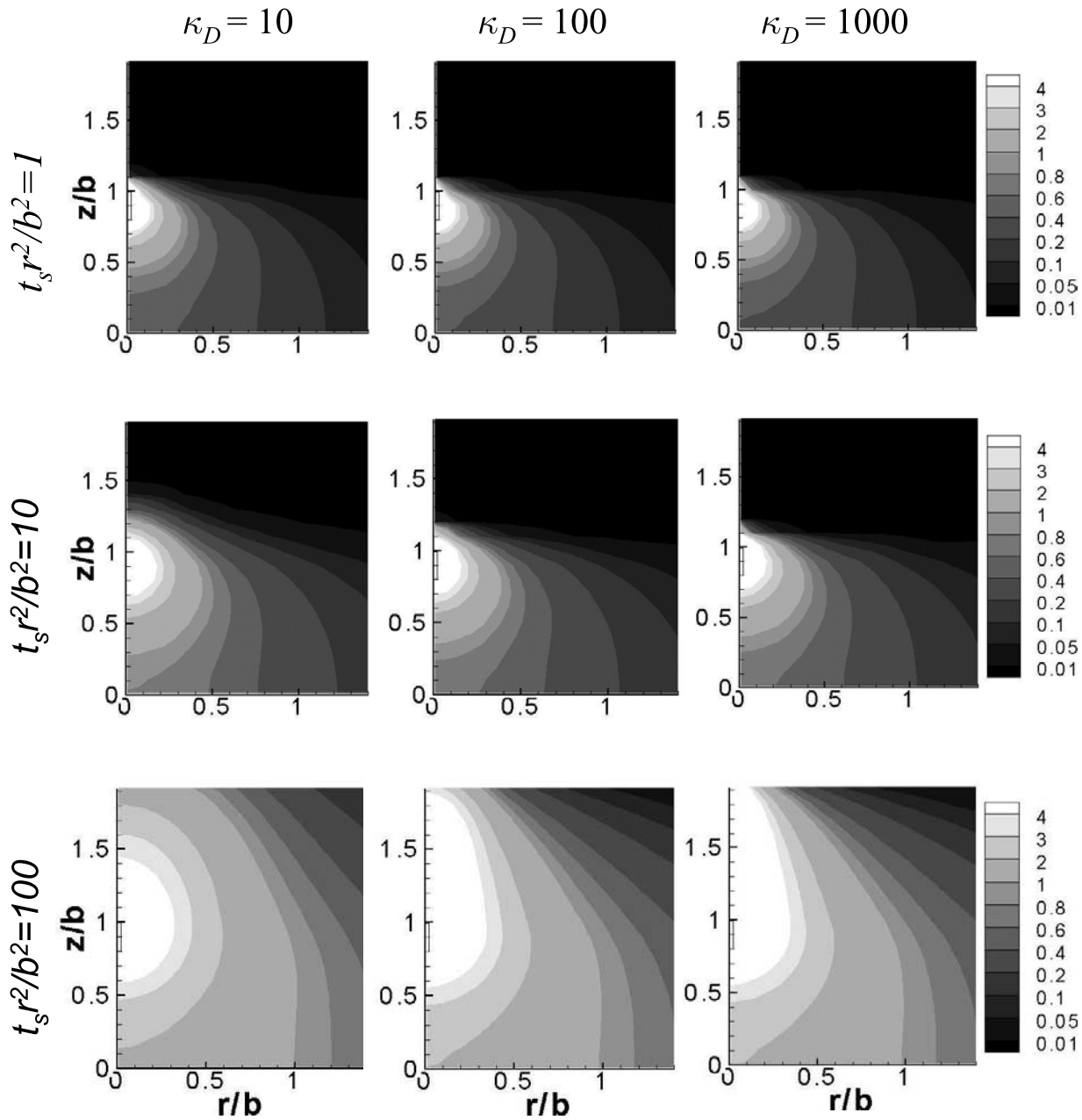


Figure 7. Contours of dimensionless drawdown $s_D = 4\pi Ts/Q$ in saturated and $\sigma_D = 4\pi T\sigma/Q$ in unsaturated zones around a pumping well penetrating the upper 20% of the saturated zone of an aquifer characterized by $K_z/K_r = 1$, $S/S_y = 0.01$, $\kappa_D = 10, 100$, and 1000 at dimensionless times $t_s r^2/b^2 = 1, 10$, and 100 .

urated flow effect significantly due to a reduction in vertical flux components relative to horizontal components.

[22] Finally, we plot in Figure 7 contours of dimensionless drawdown $s_D = 4\pi Ts/Q$ in the saturated zone, and $\sigma_D = 4\pi T\sigma/Q$ in the unsaturated zone, around a partially penetrating pumping well at $t_s r^2/b^2 = 1, 10$ and 100 for $\kappa_D = 10, 100$ and 1000 . Figure 7 reveals with clarity the three-dimensional nature of flow around the pumping well in both the saturated and the unsaturated domains. The effect of pumping is seen to propagate into the unsaturated zone ($z/b > 1$) at a rate that increases with decreasing κ_D everywhere except in a small region near the pumping well. As κ_D increases, the unsaturated zone provides less water to

the well, causing drawdown in the saturated zone to increase. Even if pressure gradients increase with increasing κ_D , fluxes in the unsaturated zone decrease because unsaturated hydraulic conductivity diminishes exponentially with this parameter.

4. Analysis of Cape Cod Pumping Test Data

[23] We consider an aquifer test conducted by the US Geological Survey in a Glacial Outwash Deposit at Cape Cod, Massachusetts [Moench *et al.*, 2001]. The aquifer, composed mainly of sand and gravel deposits estimated to be 51.82 m (170 feet) thick, is bounded from below by fine-

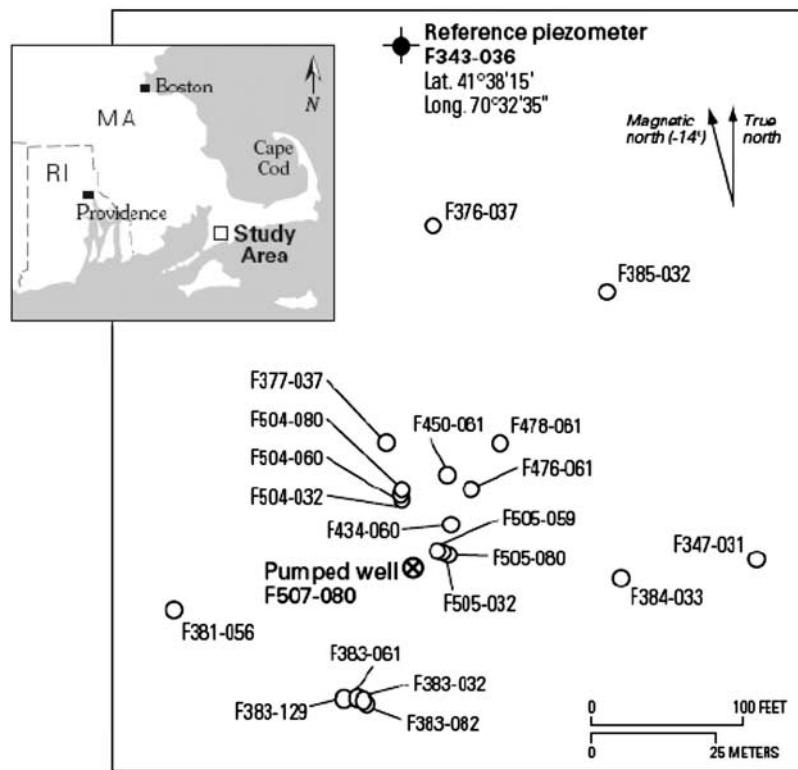


Figure 8. Regional location and local plan view of pumping well (F507-080), piezometers, and observation wells at Cape Cod study site. After Moench *et al.* [2001].

grained material of relatively low permeability [LeBlanc, 1984; LeBlanc *et al.*, 1986; Masterson *et al.*, 1997]. A well screened 4.02 m (13.2 feet) to 18.35 m (60.2 feet) below the initial water table (defined here, in the absence of information about a capillary fringe, as the traditional zero pressure isobar), located approximately 5.79 m (19 feet) below land surface and 14.26 m (46.8 feet) above mean sea level, was pumped at a constant rate of $0.0202 \text{ m}^3/\text{s}$ (320 gallons per minute) for 72 hours. Hydraulic head responses were monitored in 20 piezometers and observation wells installed at diverse locations, distances and depths around the pumping well (Figures 8 and 9). Completion data for all wells and piezometers are provided in Table 1.

[24] We fitted our solution individually and simultaneously to time-drawdown data from 20 piezometers and observation wells, and simultaneously from three piezometers in each of two clusters, by minimizing the sum of squared differences between observed and computed drawdowns. The fits yielded estimates of the parameters K_r , $K_D = K_z/K_r$, S_s , S_y , κ_D and κ listed in Table 2. Figures 10–12 show optimum fits to data from a cluster of three piezometers nearest to the pumping well (at radial distances of about 6.10 m (20 feet)). Figures 13–15 show fits to data from a similar cluster at a larger radial distance (of about 15.24 m (50 feet)) from the pumping well. Figure 16 does so for an observation well at a radial distance of about 20.12 m (66 feet). Figure 17 does so for the deepest piezometer (at depth 32.86 m (107.8 feet)) at a radial distance of about 29.57 m (97 feet), and Figure 18 does so for a shallow piezometer at the farthest radial distance of about 68.58 m (225 feet); fits to data from the remaining

piezometers and wells are given by Tartakovsky [2006]. We deem our simultaneous fits, obtained using a single physically based unsaturated zone parameter κ_D , to be the most reliable and to constitute an improvement over those reported previously using three empirical parameters α_1 , α_2 , α_3 by Moench *et al.* [2001] and Moench [2004].

[25] Table 2 lists all parameter estimates obtained by fitting our analytical solution simultaneously to all 20 sets of available time-drawdown data simultaneously (first row), to three piezometers in each of two clusters (second and third rows), and individually to each of the 20 data sets (remaining rows). Table 2 also lists the corresponding root mean square residual errors (average square differences between computed and recorded drawdowns). As one might expect, the results of individual fits exhibit some variability: K_r fluctuates between 1.02×10^{-3} and $1.78 \times 10^{-3} \text{ m/s}$ (0.20 and 0.34 feet/min), K_D between 0.2 and 1.1, S_s between 3.28×10^{-5} and $2.30 \times 10^{-4} \text{ m}^{-1}$ (1.00×10^{-5} and $7.01 \times 10^{-5} \text{ feet}^{-1}$), S_y between 0.10 and 0.38, and κ between 0.156 and 0.349 m^{-1} (0.0475 and 0.106 feet^{-1}). The variability is due in part to estimation errors and in part to spatial variability of medium properties within this mildly heterogeneous aquifer; Hess *et al.* [1992] estimated the variance of log conductivity in the sand and gravel sediments to be 0.24 with a vertical correlation scale of 0.189–0.381 m (0.620–1.250 feet). Parameter estimates obtained by simultaneous fits to all 20 time-drawdown data sets, and to 3 sets in each of two clusters of piezometers, are well within the range of those obtained by individual fits.

[26] It is worth noting that the pumping rate $Q = 320 \text{ gal/min} = 0.0202 \text{ m}^3/\text{s}$ is much smaller than $K_r b^2 = 1.02 \times$

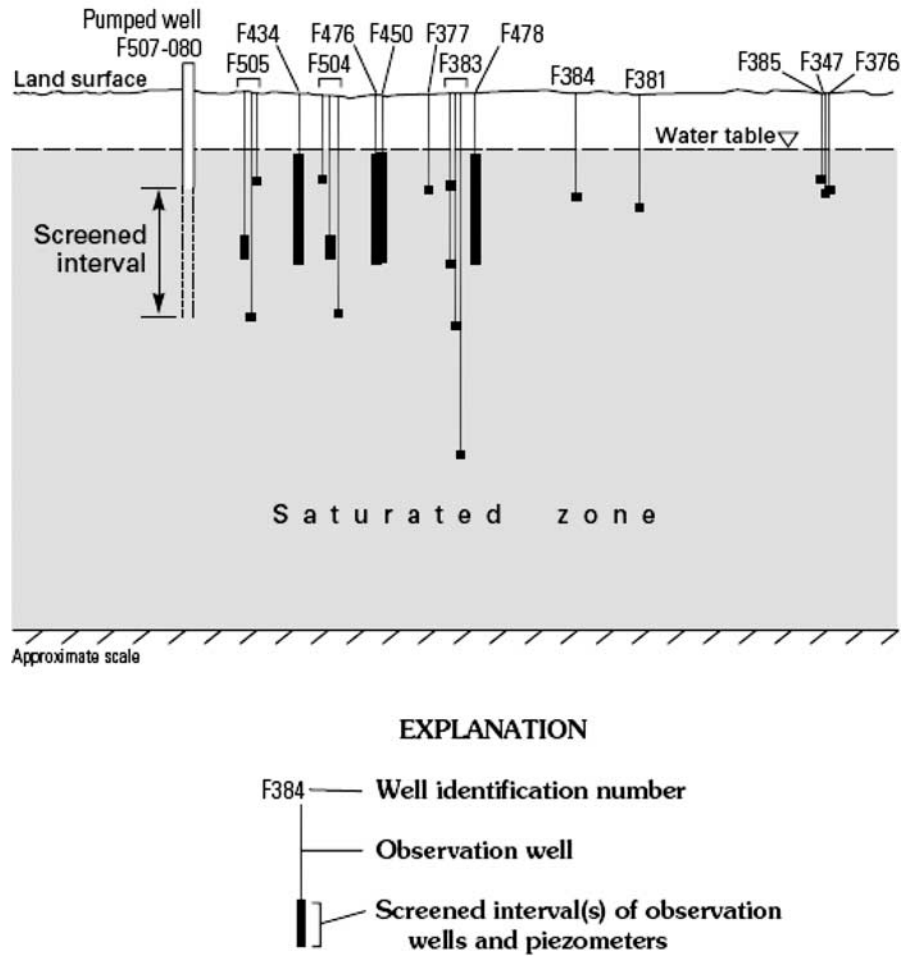


Figure 9. Vertical cross section of aquifer at Cape Cod study site showing lengths and positions of some piezometers and observation wells. After *Moench et al.* [2001].

10^{-3} m/s (0.20 feet/min) $\times 18.349^2$ m² (60.2² feet²) = 0.342 m³/s (724.8 feet³/min), satisfying the criterion under which we took our solution to be valid.

[27] Table 3 compares parameter estimates obtained by fitting our solution simultaneously to all 20 time-drawdown data sets and those obtained by *Moench et al.* [2001] and *Moench* [2004] upon fitting their solution simultaneously to these same data. Whereas our estimates of K_r (1.02×10^{-3} m/s or 0.201 feet/min) and K_z (8.13×10^{-4} m/s or 0.160 feet/min) are quite similar to those of the above authors, our estimate of S_y (0.18) is smaller and that of S_s (9.84×10^{-5} m⁻¹ or 3.00×10^{-5} feet⁻¹) larger than theirs (0.26 and 4.27×10^{-5} m⁻¹ or 1.30×10^{-5} feet⁻¹, respectively). Our estimate of κ (0.156 m⁻¹ or 0.0475 feet⁻¹) is not comparable to their empirical parameters α_1 , α_2 , α_3 , which are therefore not listed. Our estimate of 8.1 for κ_D is close to the value of 10 above which, according to our theory, the impact of unsaturated flow on drawdown at some depth below the water table becomes relatively minor.

[28] We note that since the solution of *Kroszynski and Dagan* [1975] exhibits a jump at time zero, it would be unable to reproduce the early rising limb of any of the observed time-drawdown behaviors. Graphical matches of

Table 1. Cape Cod Well and Piezometer Completion Data^a

Designation	Radial Distance r , m/feet	Depth d , m/feet	Screen Length $l - d$, m/feet
F507-080	0.102/0.333	4.023/13.2	14.326/47
F505-032	7.285/23.9	3.261/10.7	0.610/2
F505-059	5.944/19.5	9.327/30.6	2.743/9
F505-080	6.584/21.6	17.800/58.4	0.610/2
F504-032	14.204/46.6	2.926/9.6	0.610/2
F504-060	15.179/49.8	9.144/30.0	2.743/9
F504-080	16.185/53.1	17.526/57.5	0.610/2
F377-037	25.939/85.1	4.054/13.3	0.610/2
F383-032	28.346/93.0	3.6884/12.1	0.610/2
F383-061	28.316/92.9	12.162/39.9	0.610/2
F383-082	28.895/94.8	18.837/61.8	0.610/2
F383-129	29.474/96.7	32.858/107.8	0.610/2
F384-033	41.849/137.3	4.816/15.8	0.610/2
F381-056	48.707/159.8	6.096/20.0	0.610/2
F347-031	68.793/225.7	4.511/14.8	0.610/2
F434-060	11.765/38.6	0.610/2.0	11.887/39
F450-061	20.208/66.3	0.518/1.7	11.887/39
F476-061	19.995/65.6	0.671/2.2	11.887/39
F478-061	30.876/101.3	0.671/2.2	11.887/39
F385-032	68.458/224.6	3.048/10.0	0.610/2
F376-037	69.373/227.6	4.023/13.2	0.610/2

^aAfter *Moench et al.* [2001].

Table 2. Parameter Estimates and Root-Mean-Square Residual Drawdown Errors

Solution Fitted to	K_r , m/s	K_z/K_r	S_s , m^{-1}	S_y	$\kappa_D = \kappa b$	κ , m^{-1}	Error, m
All data	1.02×10^{-3}	0.8	9.84×10^{-5}	0.18	8.1	0.156	0.350
F505 Cluster	1.02×10^{-3}	0.9	9.84×10^{-5}	0.18	10.1	0.195	0.582
F504 Cluster	1.27×10^{-3}	0.5	9.84×10^{-5}	0.10	18.1	0.349	0.253
F505-032	1.02×10^{-3}	0.7	1.31×10^{-4}	0.18	12.1	0.234	0.126
F505-059	1.02×10^{-3}	0.8	9.84×10^{-5}	0.28	14.1	0.272	0.296
F505-080	1.02×10^{-3}	0.8	1.31×10^{-4}	0.38	18.1	0.349	0.302
F504-032	1.02×10^{-3}	0.6	9.84×10^{-5}	0.15	10.1	0.195	0.041
F504-060	1.02×10^{-3}	0.7	6.56×10^{-5}	0.15	8.1	0.156	0.125
F504-080	1.27×10^{-3}	0.5	9.84×10^{-5}	0.13	18.1	0.349	0.080
F383-061	1.02×10^{-3}	0.6	9.84×10^{-5}	0.22	10.1	0.195	0.031
F434-060	1.02×10^{-3}	1.1	1.64×10^{-4}	0.16	18.1	0.349	0.108
F376-037	1.78×10^{-3}	0.2	3.28×10^{-5}	0.13	16.1	0.311	0.044
F450-061	1.02×10^{-3}	0.8	1.64×10^{-4}	0.16	8.1	0.156	0.045
F385-032	1.52×10^{-3}	0.2	1.64×10^{-4}	0.18	14.1	0.272	0.019
F377-037	1.02×10^{-3}	0.7	6.56×10^{-5}	0.19	8.1	0.156	0.024
F383-032	1.02×10^{-3}	0.6	2.30×10^{-4}	0.25	8.1	0.156	0.017
F383-082	1.02×10^{-3}	0.5	1.64×10^{-4}	0.22	8.1	0.156	0.020
F383-129	1.27×10^{-3}	0.4	1.31×10^{-4}	0.13	18.1	0.349	0.009
F384-033	1.27×10^{-3}	0.3	6.56×10^{-5}	0.19	18.1	0.349	0.027
F381-056	1.27×10^{-3}	0.3	9.84×10^{-5}	0.15	10.1	0.195	0.020
F476-061	1.02×10^{-3}	0.8	1.64×10^{-4}	0.16	8.1	0.156	0.088
F478-061	1.27×10^{-3}	0.5	1.31×10^{-4}	0.16	18.1	0.349	0.024
F347-031	1.27×10^{-3}	0.3	6.56×10^{-5}	0.19	16.1	0.311	0.005

Neuman's [1972, 1974] solution to the Cape Cod data and corresponding aquifer parameter estimates are given by Moench *et al.* [2001].

5. Summary and Findings

[29] A three-dimensional analytical solution with axial symmetry was developed for drawdown in an unconfined aquifer due to pumping at a constant rate from a partially penetrating well. Our solution accounts for aquifer com-

pressibility in a manner similar to that of Neuman [1974]. It describes flow above the water table by a linearized three-dimensional version of Richards' equation in which hydraulic conductivity and water content vary exponentially with incremental capillary pressure head relative to its air entry value in a manner similar to that of Kroszynski and Dagan [1975]. The water table is thus defined as an isobar at a nonpositive air entry pressure, overlying a saturated capillary fringe within which pressures are likewise non-

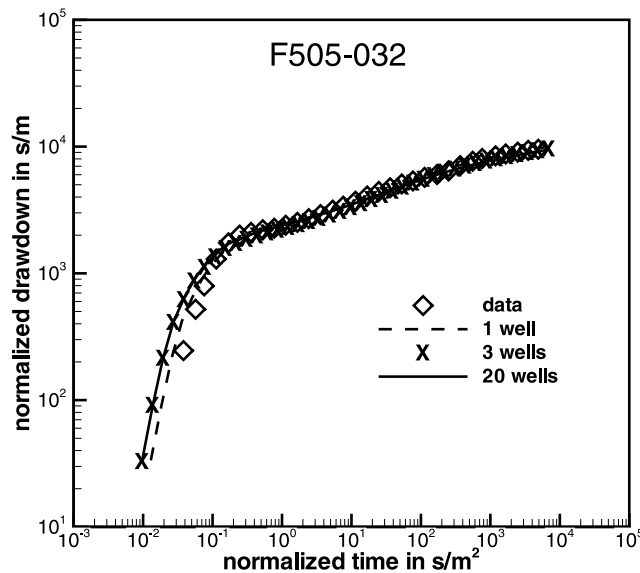


Figure 10. Normalized drawdown $s_n = 4\pi bs/Q$ versus normalized time $t_n = t/r^2$ in piezometer F505-032. Diamonds represent data from Moench *et al.* [2001]. Dashes, crosses, and solid curve represent analytical solution fitted to these data, to data from all three piezometers in the F505 cluster, and to data from all 20 piezometers and observations wells, respectively.

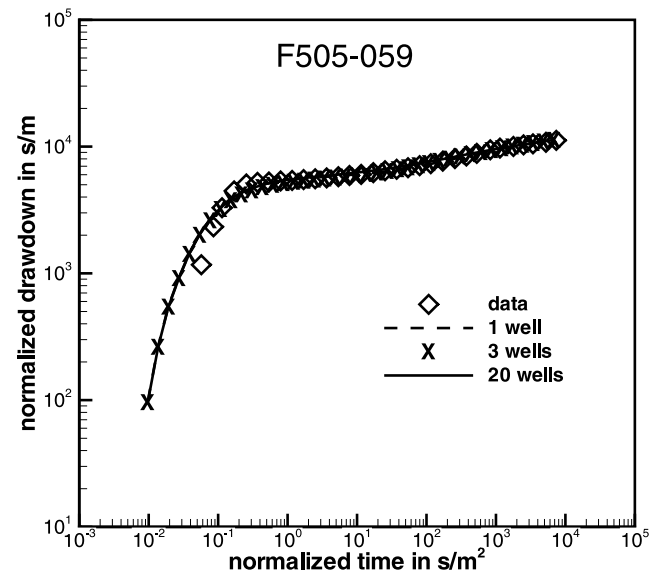


Figure 11. Normalized drawdown $s_n = 4\pi bs/Q$ versus normalized time $t_n = t/r^2$ in piezometer F505-059. Diamonds represent data from Moench *et al.* [2001]. Dashes, crosses, and solid curve represent analytical solution fitted to these data, to data from all three piezometers in the F505 cluster, and to data from all 20 piezometers and observations wells, respectively.

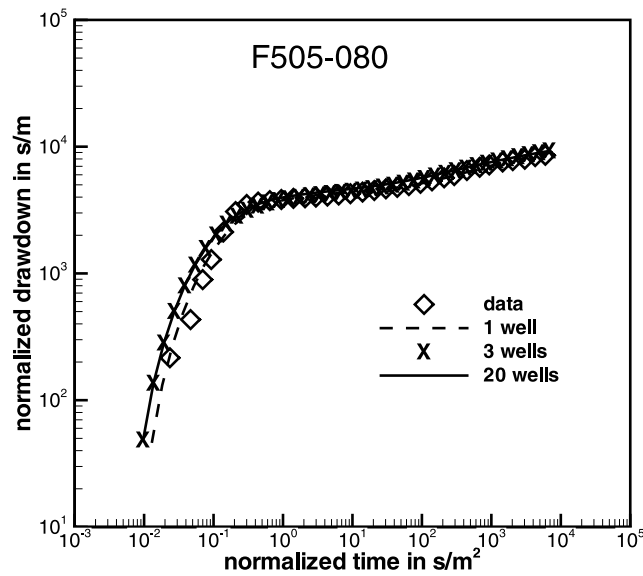


Figure 12. Normalized drawdown $s_n = 4\pi bs/Q$ versus normalized time $t_n = t/r^2$ in piezometer F505-080. Diamonds represent data from Moench *et al.* [2001]. Dashes, crosses, and solid curve represent analytical solution fitted to these data, to data from all three piezometers in the F505 cluster, and to data from all 20 piezometers and observations wells, respectively.

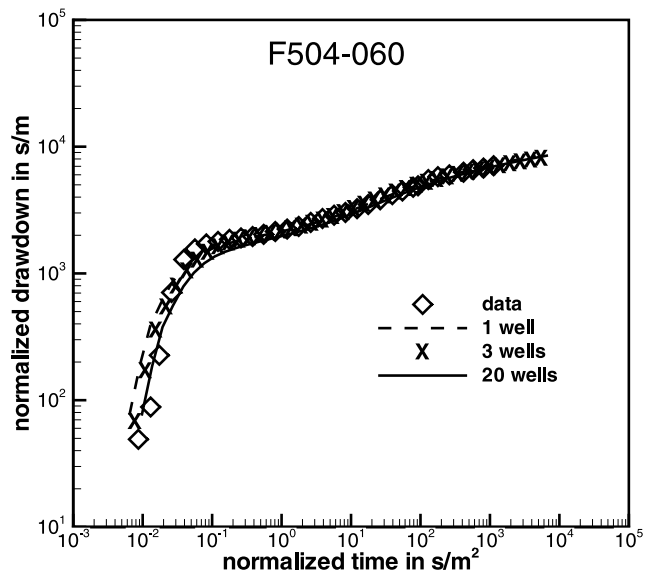


Figure 14. Normalized drawdown $s_n = 4\pi bs/Q$ versus normalized time $t_n = t/r^2$ in piezometer F504-060. Diamonds represent data from Moench *et al.* [2001]. Dashes, crosses, and solid curve represent analytical solution fitted to these data, to data from all three piezometers in the F504 cluster, and to data from all 20 piezometers and observations wells, respectively.

positive. Unsaturated and saturated flows are fully coupled. Unsaturated constitutive soil properties are characterized by an exponent κ having the dimension of inverse length, or equivalently a dimensionless exponent $\kappa_D = \kappa b$ where b is the initial saturated thickness. Part of our solution is given

in the time domain and part in Laplace transformed domain. The latter part was inverted back into the time domain using a numerical algorithm. We found the following.

[30] 1. Unsaturated flow has significant impact on dimensionless log-log time-drawdown behavior in the saturated zone when κ_D is less than 1 (the aquifer has large retention

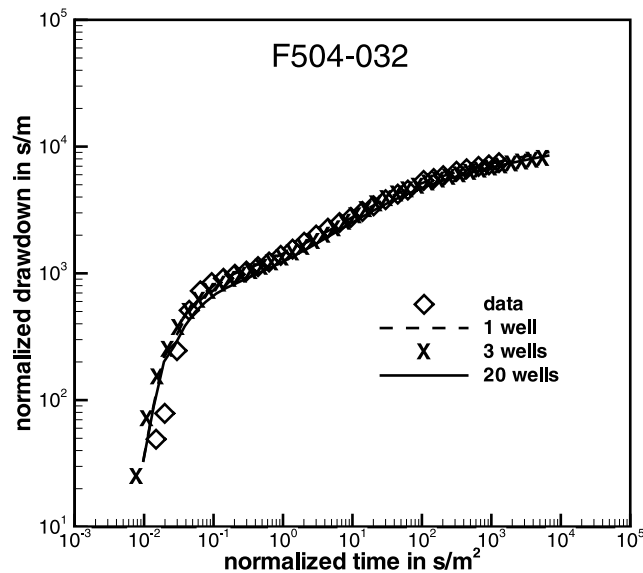


Figure 13. Normalized drawdown $s_n = 4\pi bs/Q$ versus normalized time $t_n = t/r^2$ in piezometer F504-032. Diamonds represent data from Moench *et al.* [2001]. Dashes, crosses, and solid curve represent analytical solution fitted to these data, to data from all three piezometers in the F504 cluster, and to data from all 20 piezometers and observations wells, respectively.

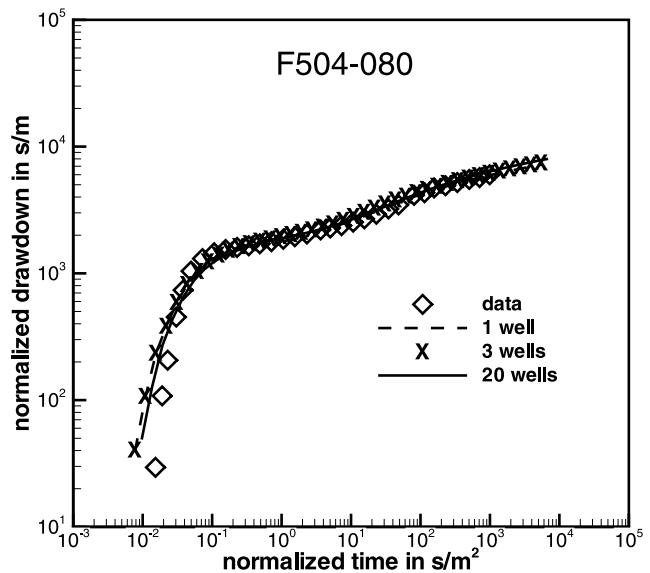


Figure 15. Normalized drawdown $s_n = 4\pi bs/Q$ versus normalized time $t_n = t/r^2$ in piezometer F504-080. Diamonds represent data from Moench *et al.* [2001]. Dashes, crosses, and solid curve represent analytical solution fitted to these data, to data from all three piezometers in the F504 cluster, and to data from all 20 piezometers and observations wells, respectively.

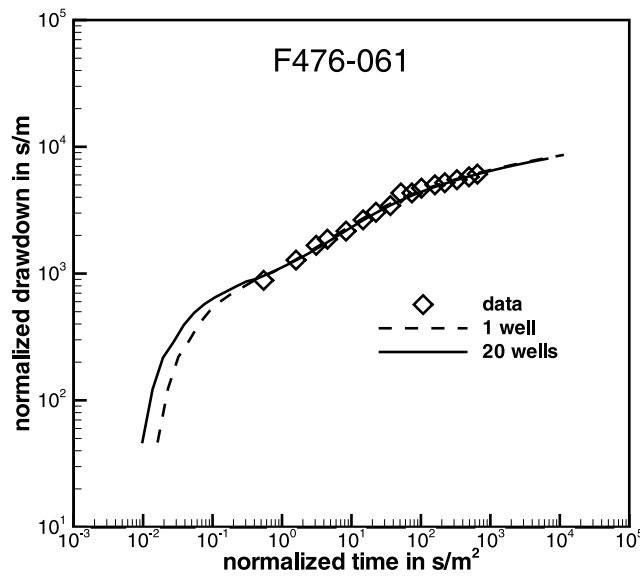


Figure 16. Normalized drawdown $s_n = 4\pi bs/Q$ versus normalized time $t_n = t/r^2$ in observation well F476-061. Diamonds represent data from *Moench et al.* [2001]. Dashes and solid curve represent analytical solution fitted to these data and to data from all 20 piezometers and observations wells, respectively.

capacity and/or small initial saturated thickness). The impact of unsaturated on saturated flow everywhere but in the immediate vicinity of the water table diminishes as κ_D increases, becoming small or insignificant at κ_D greater than 10 (small retention capacity and/or large saturated thickness). At these large values of κ_D our solution coincides with that of *Neuman* [1974].

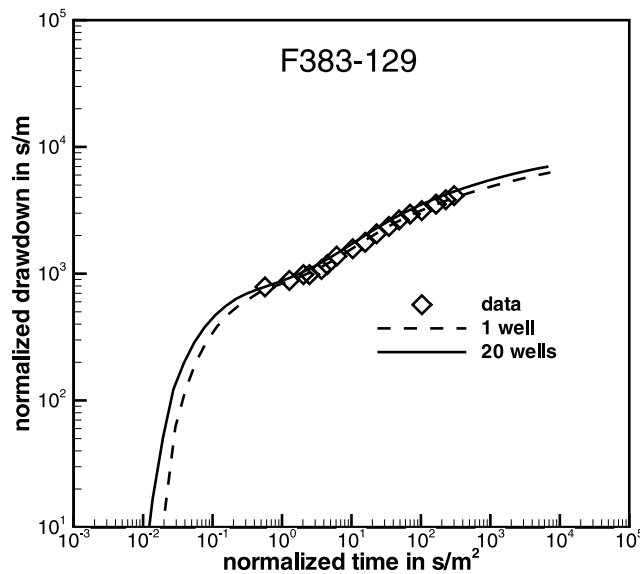


Figure 17. Normalized drawdown $s_n = 4\pi bs/Q$ versus normalized time $t_n = t/r^2$ in piezometer F383-129. Diamonds represent data from *Moench et al.* [2001]. Dashes and solid curve represent analytical solution fitted to these data and to data from all 20 piezometers and observations wells, respectively.

[31] 2. When κ_D is small, unsaturated flow causes dimensionless drawdown in the saturated zone to increase at early and intermediate dimensionless times and to decrease at late time, the impact being least pronounced at early time and most pronounced at intermediate time. The overall effect is to eliminate S-shaped inflections, rendering the dimensionless log-log time-drawdown curve more and more monotonic. Whereas S-shaped behavior at large κ_D reflects the dominance of artesian (compressive internal) storage during early time and of near instantaneous drainage at the water table during late time, monotonic behavior at small κ_D reflects the dominance of slow drainage from the unsaturated zone.

[32] 3. The effect of unsaturated flow on drawdown in a piezometer or an observation well, completed within the saturated zone, diminishes as the pumping or the observation well becomes more fully penetrating. As the pumping well penetrates the saturated zone more fully vertical flux components within this zone, which dominate drainage from the overlying unsaturated zone, diminish. As an observation well penetrates the aquifer more fully, the vertically averaged drawdown one measures in it becomes less sensitive to the effect of vertical hydraulic gradients around it, and thus to drainage from the unsaturated zone.

[33] 4. The effect of unsaturated flow on drawdown in the saturated zone is most pronounced at and just below the water table, more so closer to the pumping well than farther from it, and least near the aquifer bottom. A decrease in κ_D generally leads to an increase in point drawdown.

[34] 5. As horizontal saturated hydraulic conductivity increases relative to the vertical, the effect of unsaturated flow on drawdown in the saturated zone diminishes, and the artesian effect of aquifer compressibility becomes more pronounced, due to the enhancement of horizontal fluxes relative to vertical fluxes.

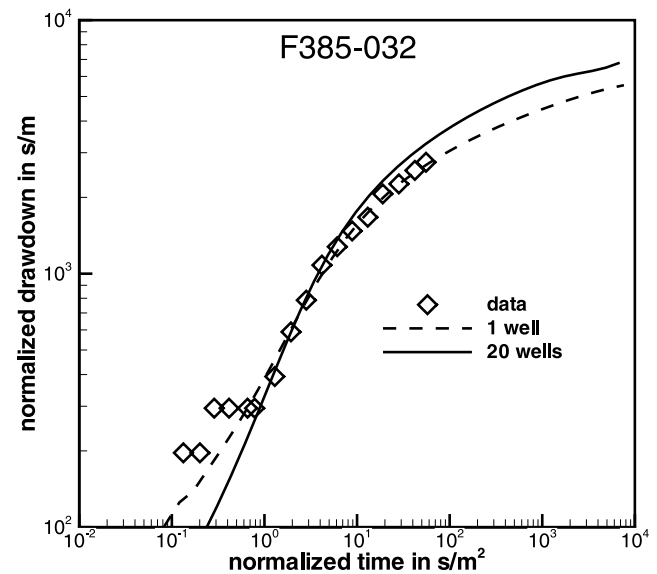


Figure 18. Normalized drawdown $s_n = 4\pi bs/Q$ versus normalized time $t_n = t/r^2$ in piezometer F385-032. Diamonds represent data from *Moench et al.* [2001]. Dashes and solid curve represent analytical solution fitted to these data and to data from all 20 piezometers and observations wells, respectively.

Table 3. Parameter Estimates Obtained by Fitting Our Solution to All 20 Cape Cod Time-Drawdown Records Compared With Corresponding Estimates by *Moench et al.* [2001]

	K_r , m/s (feet/min)	K_z , m/s (feet/min)	S_s , m ⁻¹ (feet ⁻¹)	S_y	$\kappa_D = \kappa b$	κ , m ⁻¹
Moench et al.	1.17×10^{-3} (0.230)	7.11×10^{-4} (0.140)	4.27×10^{-5} (1.30×10^{-5})	0.26	-	-
This study	1.02×10^{-3} (0.201)	8.13×10^{-4} (0.160)	9.84×10^{-5} (3.00×10^{-5})	0.18	8.1	0.156

[35] 6. Flow around the pumping well is three-dimensional, with axial symmetry, in both the saturated and the unsaturated zones. The effect of pumping propagates vertically and horizontally through the unsaturated zone at a rate that generally increases with decreasing κ_D . As κ_D increases, the unsaturated zone provides less water to the well, causing drawdown in the saturated zone to increase. Even if pressure gradients increase with increasing κ_D , fluxes in the unsaturated zone decrease because unsaturated hydraulic conductivity diminishes exponentially with this parameter.

[36] 7. We fitted our solution individually and simultaneously to time-drawdown data from 20 piezometers and observation wells, and simultaneously to data from three piezometers in each of two clusters, obtained during an aquifer test in a Glacial Outwash Deposit at Cape Cod, Massachusetts, by the US Geological Survey [*Moench et al.*, 2001]. Optimum fits were obtained by minimizing the sum of squared differences between observed and computed drawdowns. The results of individual fits exhibit some variability due in part to estimation errors and in part to spatial variability of medium properties. Parameter estimates obtained by simultaneous fits to all or part of the data are well within the range of these variations.

[37] 8. Our simultaneous fits to all 20 time-drawdown records, obtained using a single physically based unsaturated zone parameter κ_D , are an improvement over those reported previously using three empirical parameters by *Moench et al.* [2001] and *Moench* [2004]. This is true even though these authors have accounted for storage in the pumping well and we have not, implying that such storage has little impact on test interpretation at the Cape Cod site. Our estimates of horizontal (1.02×10^{-3} m/s or 0.201 feet/min) and vertical (8.13×10^{-4} m/s or 0.160 feet/min) saturated hydraulic conductivity are close to theirs; our estimate of specific yield (0.18) is smaller and that of specific storage (9.84×10^{-5} m⁻¹ or 3.00×10^{-5} feet⁻¹) larger than theirs (0.26 and 4.27×10^{-5} m⁻¹ or 1.30×10^{-5} feet⁻¹, respectively); but our estimate of the constitutive exponent κ (0.156 m⁻¹ or 0.0475 feet⁻¹) is not comparable to their empirical parameters.

[38] 9. Though the exponential model we use to represent unsaturated soil properties is relatively crude, it has the advantage of being parsimonious (by relying on a single parameter κ) while capturing with reasonable accuracy observed time-drawdown behavior at Cape Cod and, we hope, elsewhere.

Appendix A: Decomposition of Saturated Zone Solution

[39] In a manner analogous to *Neuman* [1974] we decompose s into two parts,

$$s = s_H + s_U \quad (\text{A1})$$

where s_H is *Hantush's* [1964] solution for a partially penetrating well in a confined aquifer, satisfying

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial s_H}{\partial r} \right) + K_D \frac{\partial^2 s_H}{\partial z^2} = \frac{1}{\alpha_s} \frac{\partial s_H}{\partial t} \quad 0 \leq z < b \quad (\text{A2})$$

$$s_H(r, z, 0) = 0 \quad (\text{A3})$$

$$s_H(\infty, z, t) = 0 \quad (\text{A4})$$

$$\frac{\partial s_H}{\partial z} = 0 \quad z = 0 \quad \text{and} \quad z = b \quad (\text{A5})$$

$$\lim_{r \rightarrow 0} \frac{\partial s_H}{\partial r} = 0 \quad 0 \leq z \leq b - l \quad b - d \leq z \leq b \quad (\text{A6})$$

$$\lim_{r \rightarrow 0} r \frac{\partial s_H}{\partial r} = -\frac{Q}{2\pi K_r(l-d)} \quad b - l \leq z \leq b - d. \quad (\text{A7})$$

and s_U is a correction due to saturated-unsaturated unconfined conditions, satisfying

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial s_U}{\partial r} \right) + K_D \frac{\partial^2 s_U}{\partial z^2} = \frac{1}{\alpha_s} \frac{\partial s_U}{\partial t} \quad 0 \leq z < b \quad (\text{A8})$$

$$s_U(r, z, 0) = 0 \quad (\text{A9})$$

$$s_U(\infty, z, t) = 0 \quad (\text{A10})$$

$$\frac{\partial s_U}{\partial z} = 0 \quad z = 0 \quad (\text{A11})$$

$$\lim_{r \rightarrow 0} \frac{\partial s_U}{\partial r} = 0 \quad 0 \leq z \leq b \quad (\text{A12})$$

subject to interface conditions (29)–(30) written as

$$s_H + s_U - \sigma = 0 \quad z = b \quad (\text{A13})$$

$$\frac{\partial s_H}{\partial z} + \frac{\partial s_U}{\partial z} - \frac{\partial \sigma}{\partial z} = 0 \quad z = b. \quad (\text{A14})$$

Appendix B: Laplace and Hankel Transformations

[40] Equations (A1)–(A14) are solved through sequential application of the Hankel transform

$$\langle f(a) \rangle = \int_0^\infty r J_0(ar) f(r) dr \quad (\text{B1})$$

and Laplace transform

$$\bar{f}(p) = \int_0^\infty f(t)e^{-pt} dt \quad (B2)$$

with parameters a and p , J_0 being zero-order Bessel function of the first kind.

[41] Neuman [1974, p. 309] showed that the double (Laplace and Hankel) transform of (33) and (34) is

$$\bar{s}_r = \frac{Q}{2\pi TK_D p \eta^2} \quad (B3)$$

$$\Delta \bar{s} = \frac{Q}{2\pi TK_D p} \frac{2}{\pi(l_D - d_D)} \sum_{n=1}^{\infty} \frac{1}{n} \cos[n\pi(1 - z_D)] \cdot [\sin(n\pi l_D) - \sin(n\pi d_D)] \frac{1}{\eta^2 + n^2\pi^2/b^2} \quad (B4)$$

respectively where $\eta^2 = (a^2/K_D) + (p/\alpha_s K_D)$. The Laplace transform of (23)–(28) is

$$k_0(z) \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \bar{\sigma}}{\partial r} \right) + K_D \frac{\partial}{\partial z} \left(k_0(z) \frac{\partial \bar{\sigma}}{\partial z} \right) = p \frac{C_0(z)}{K_r} \bar{\sigma} \quad b < z < b + L \quad (B5)$$

$$\bar{\sigma}(\infty, z, p) = 0 \quad (B6)$$

$$\frac{\partial \bar{\sigma}}{\partial z} = 0 \quad z = b + L \quad (B7)$$

$$\lim_{r \rightarrow 0} r \frac{\partial \bar{\sigma}}{\partial r} = 0 \quad b < z < b + L. \quad (B8)$$

The Laplace transform of (A8)–(A14) is

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \bar{s}_U}{\partial r} \right) + K_D \frac{\partial^2 \bar{s}_U}{\partial z^2} = \frac{p}{\alpha_s} \bar{s}_U \quad 0 \leq z < b \quad (B9)$$

$$\bar{s}_U(\infty, z, t) = 0 \quad (B10)$$

$$\frac{\partial \bar{s}_U}{\partial z} = 0 \quad z = 0 \quad (B11)$$

$$\lim_{r \rightarrow 0} r \frac{\partial \bar{s}_U}{\partial r} = 0 \quad 0 \leq z \leq b \quad (B12)$$

$$\bar{s}_H + \bar{s}_U - \bar{\sigma} = 0 \quad z = b \quad (B13)$$

$$\frac{\partial \bar{s}_H}{\partial z} + \frac{\partial \bar{s}_U}{\partial z} - \frac{\partial \bar{\sigma}}{\partial z} = 0 \quad z = b \quad (B14)$$

[42] Substituting the constitutive relations $k_0(z) = e^{\kappa(b-z)}$ and $C_0(z) = S_y \kappa e^{\kappa(b-z)}$ into (B5) allows rewriting it as

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \bar{\sigma}}{\partial r} \right) + K_D \frac{\partial^2 \bar{\sigma}}{\partial z^2} - K_D \kappa \frac{\partial \bar{\sigma}}{\partial z} = p \frac{S_y \kappa}{K_r} \bar{\sigma} \quad b < z < b + L \quad (B15)$$

Taking the Hankel transform of (B6)–(B15), setting $L \rightarrow \infty$ and using the relationship [e.g., Neuman, 1974]

$$H \left\{ \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \bar{s}}{\partial r} \right) \right\} = -a^2 \bar{s} - \lim_{r \rightarrow 0} r \frac{\partial \bar{s}}{\partial r} \quad (B16)$$

leads to

$$-a^2 \bar{\sigma} + K_D \frac{\partial^2 \bar{\sigma}}{\partial z^2} - K_D \kappa \frac{\partial \bar{\sigma}}{\partial z} = p \frac{\kappa_D}{\alpha_y} \bar{\sigma}, \quad b < z < \infty \quad (B17)$$

$$\frac{\partial \bar{\sigma}}{\partial z} = 0, \quad z = \infty \quad (B18)$$

$$-a^2 \bar{s}_U + K_D \frac{\partial^2 \bar{s}_U}{\partial z^2} = \frac{p}{\alpha_s} \bar{s}_U \quad 0 \leq z < b \quad (B19)$$

$$\frac{\partial \bar{s}_U}{\partial z} = 0 \quad z = 0 \quad (B20)$$

$$\bar{s}_H + \bar{s}_U - \bar{\sigma} = 0 \quad z = b \quad (B21)$$

$$\frac{\partial \bar{s}_H}{\partial z} + \frac{\partial \bar{s}_U}{\partial z} - \frac{\partial \bar{\sigma}}{\partial z} = 0 \quad z = b. \quad (B22)$$

The solution of (B17) subject to (B18) and (B19) is

$$\bar{\sigma} = \left(\frac{d\bar{s}_H}{dz} + \frac{d\bar{s}_U}{dz} \right)_{z=b} \exp \left[\left(\frac{\kappa}{2} - \sqrt{\frac{\kappa^2}{4} + \lambda^2} \right) (z - b) \right] \cdot \left(\frac{\kappa}{2} - \sqrt{\frac{\kappa^2}{4} + \lambda^2} \right)^{-1} \quad (B23)$$

where $\lambda^2 = \frac{a^2}{K_D} + \frac{\kappa_D}{\alpha_y K_D} p$. A general solution of (B19) subject to (B20) is

$$\bar{s}_U = \rho \cosh(\eta z) \quad (B24)$$

where $\eta^2 = (a^2/K_D) + (p/\alpha_s K_D)$ and ρ is determined on the basis of (B21), yielding

$$\bar{s}_U = \cosh(\eta z) \frac{\left. \frac{d\bar{s}_H}{dz} \right|_{z=b} \left(\frac{\kappa}{2} - \sqrt{\frac{\kappa^2}{4} + \lambda^2} \right)^{-1} - \bar{s}_H|_{z=b}}{\cosh(\eta b) - \eta \sinh(\eta b) \left(\frac{\kappa}{2} - \sqrt{\frac{\kappa^2}{4} + \lambda^2} \right)^{-1}}. \quad (B25)$$

From (B3)–(B4) it follows that

$$\bar{s}_H|_{z=b} = \frac{Q}{2\pi TK_D p} \left\{ \frac{1}{\eta^2} + \frac{2}{\pi(l_D - d_D)} \sum_{n=1}^{\infty} \frac{1}{n} \cdot [\sin(n\pi l_D) - \sin(n\pi d_D)] \frac{1}{\eta^2 + n^2\pi^2/b^2} \right\} \quad (B26)$$

$$\left. \frac{d\bar{s}_H}{dz} \right|_{z=b} = 0 \quad (B27)$$

From Neuman [1974, p. 309] it follows that the infinite sum in (B26) can be represented in closed form as

$$\begin{aligned} & \frac{2}{\pi(l_D - d_D)} \sum_{n=1}^{\infty} \frac{1}{n} [\sin(n\pi l_D) - \sin(n\pi d_D)] \frac{1}{\eta^2 + n^2\pi^2/b^2} \\ &= \frac{2}{\pi(l_D - d_D)} \frac{1}{\eta^2} \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{n}{n^2 + b^2\eta^2/\pi^2} \right) \\ & \quad \cdot [\sin(n\pi l_D) - \sin(n\pi d_D)] \\ &= -\frac{1}{\eta^2} \frac{\sinh[b\eta(1 - l_D)] - \sinh[b\eta(1 - d_D)]}{\eta^2(l_D - d_D) \sinh(b\eta)} \end{aligned} \quad (B28)$$

Substituting this into (B26) yields

$$\bar{s}_H|_{z=b} = -\frac{Q}{2\pi TK_{DP}} \frac{\sinh[b\eta(1 - l_D)] - \sinh[b\eta(1 - d_D)]}{\eta^2(l_D - d_D) \sinh(b\eta)}. \quad (B29)$$

Substituting (B27) and (B29) into (B25) gives

$$\begin{aligned} \bar{s}_U &= \frac{Q}{2\pi TK_{DP}} \cosh(\eta z) \left[\cosh(\eta b) - \eta \sinh(\eta b) \left(\frac{\kappa}{2} - \sqrt{\frac{\kappa^2}{4} + \lambda^2} \right)^{-1} \right]^{-1} \\ & \quad \cdot \frac{\sinh[b\eta(1 - l_D)] - \sinh[b\eta(1 - d_D)]}{\eta^2(l_D - d_D) \sinh(b\eta)}. \end{aligned} \quad (B30)$$

Substituting (B27) and (B30) into (B23) results in

$$\begin{aligned} \bar{\sigma} &= \frac{Q\eta}{2\pi TK_{DP}} \sinh(\eta b) \left[\cosh(\eta b) - \eta \sinh(\eta b) \left(\frac{\kappa}{2} - \sqrt{\frac{\kappa^2}{4} + \lambda^2} \right)^{-1} \right]^{-1} \\ & \quad \cdot \frac{\sinh[b\eta(1 - l_D)] - \sinh[b\eta(1 - d_D)]}{\eta^2(l_D - d_D) \sinh(b\eta)} \\ & \quad \cdot \exp \left[\left(\frac{\kappa}{2} - \sqrt{\frac{\kappa^2}{4} + \lambda^2} \right) (z - b) \right] \left(\frac{\kappa}{2} - \sqrt{\frac{\kappa^2}{4} + \lambda^2} \right)^{-1} \end{aligned} \quad (B31)$$

The inverse Hankel transforms of (B30) and (B31) are

$$\begin{aligned} \bar{s}_U &= \frac{Q}{2\pi TK_{DP}} \int_0^{\infty} \cosh(\eta z) \left[\cosh(\eta b) - \eta \sinh(\eta b) \left(\frac{\kappa}{2} - \sqrt{\frac{\kappa^2}{4} + \lambda^2} \right)^{-1} \right]^{-1} \\ & \quad \cdot \frac{\sinh[b\eta(1 - l_D)] - \sinh[b\eta(1 - d_D)]}{\eta^2(l_D - d_D) \sinh(b\eta)} aJ_0(ar) da \end{aligned} \quad (B32)$$

$$\begin{aligned} \bar{\sigma} &= \frac{Q}{2\pi TK_{DP}} \int_0^{\infty} \eta \sinh(\eta b) \left[\cosh(\eta b) - \eta \sinh(\eta b) \left(\frac{\kappa}{2} - \sqrt{\frac{\kappa^2}{4} + \lambda^2} \right)^{-1} \right]^{-1} \\ & \quad \cdot \frac{\sinh[b\eta(1 - l_D)] - \sinh[b\eta(1 - d_D)]}{\eta^2(l_D - d_D) \sinh(b\eta)} \\ & \quad \cdot \exp \left[\left(\frac{\kappa}{2} - \sqrt{\frac{\kappa^2}{4} + \lambda^2} \right) (z - b) \right] \left(\frac{\kappa}{2} - \sqrt{\frac{\kappa^2}{4} + \lambda^2} \right)^{-1} aJ_0(ar) da \end{aligned} \quad (B33)$$

respectively. Defining a new variable $y = ar/\beta^{1/2}$ these transform into (35) and (36), respectively.

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